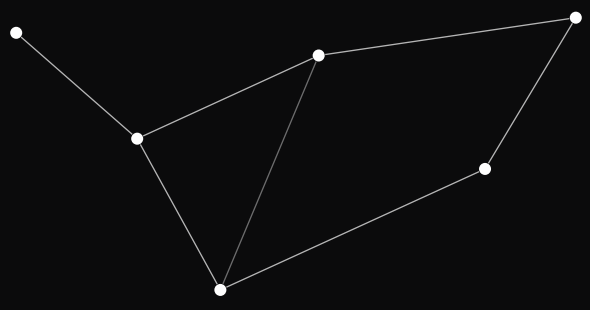
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WHITE PAPER

A fast and open-source Physical-Optics implementation for AFR/SFPB reflector antennas



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Abstract

We present an open-source Physical Optics (PO) implementation for the secondary-pattern analysis of front-fed paraboloid reflectors excited by Array-Fed Reflector (AFR) and Single-Feed-Per-Beam (SFPB) architectures, validated against GRASP to within measurement precision. The kernel is implemented once in Rust and exposed to Python and MATLAB through a common C ABI. We describe a coarse angular grid plus interpolation scheme that reduces the number of directions at which the PO integral must be evaluated by an order of magnitude at negligible accuracy cost, and report measured execution times, per-beam radiation patterns, and known limitations of the model. All code is open source.

Introduction

Multibeam satellite antennas realize their coverage either as one feed per beam (SFPB) or as a feed array combined through, for example, Conjugate Field Matching (CFM) into an Array-Fed Reflector (AFR). AFR trades feed-chain complexity for beam-shaping flexibility and graceful degradation under element failure or surface distortion [1]. Both architectures are governed by the same secondary-pattern problem: given a reflector geometry and a feed (or feed array) excitation, predict the far-field radiation pattern.

Physical Optics (PO) solves this by integrating the induced surface current $\mathbf{J}_s = 2\hat{\mathbf{n}} \times \mathbf{H}_i$ over the illuminated reflector, rather than solving the full integral equation (Method of Moments) or applying ray-based asymptotic corrections (GTD/UTD). For electrically large reflectors, from tens to hundreds of wavelengths, (and what we consider in this paper), PO is the standard trade-off: its accuracy is best precisely in the illuminated-zone co-polar pattern that dominates system link budget, and it is far cheaper than a full-wave solution. Its cost is still $O(\text{directions} \times \text{surface points})$, and on top of that every direction of a multibeam coverage map, every feed of an AFR array, and every design iteration *multiplies that cost*, which is what motivates the method to speed-up the code described in the Methodology section.

The reflector we consider is the axisymmetric paraboloid $z = (x^2 + y^2)/4F$, diameter D, focus at $(0, 0, F)$. Each feed is modeled by a $\cos^q \theta'$ pattern taper about its own boresight and a linearly ($\hat{x}$-polarized) Ludwig-3 co-polar field. The far-field co-polar response is

$$W(\hat{\mathbf{r}}) = \iint_S \left[\mathbf{J}_t(\mathbf{r}')\right] \cdot \hat{\mathbf{R}}(\theta, \varphi)\, e^{jk\hat{\mathbf{r}}\cdot\mathbf{r}'}\, dS', \tag{1}$$

where $\mathbf{J}_t = \mathbf{J}_s - (\mathbf{J}_s \cdot \hat{\mathbf{r}})\hat{\mathbf{r}}$ is the transverse current and $\hat{\mathbf{R}}$ is the Ludwig-3 co-polar unit vector. For M feeds with complex excitation weights w_m, AFR forms each beam by CFM,

$$w_m = \frac{v_m^*}{\|\mathbf{v}\|}, \qquad v_m = W_m(\hat{\mathbf{r}}_{\text{beam}}), \tag{2}$$

i.e. each feed's weight is matched to its own response at the beam axis; SFPB is the $M{=}1$ case of the same formula, where $|w|{=}1$ reduces to a pure phase rotation. Gain follows from

$$G(\hat{\mathbf{r}}) = 10 \log_{10}\left(\frac{k^2 Z_0}{8\pi P_{\text{rad}}} |W(\hat{\mathbf{r}})|^2\right) \text{ dBi}, \quad P_{\text{rad}} = \frac{\|\mathbf{w}\|^2 \pi}{Z_0(2q+1)}. \tag{3}$$

Equations (1)–(3) are the complete model; the reader is referred to Acosta [1] for their full derivation and for the array-compensation context that motivates AFR in the first place.

Known Limitations

Two categories of limitation apply: those intrinsic to the PO method itself, and those specific to this implementation. Intrinsic limitations to Physical Optics includes that PO evaluates a single reflection with a locally-tangent-plane current approximation. It omits edge diffraction (no fringe-current / GTD correction), multiple reflections, and blockage from feed struts and support structures; accuracy degrades toward the reflector rim and into deep sidelobes, where the tangent-plane approximation is weakest. These are the standard, accepted trade-offs against Method-of-Moments accuracy in exchange for tractable cost on electrically large apertures [1].

Limitations specific to this implementation are:

- **No mutual coupling.** Feed elements are excited independently. For AFR this permits the idealized aperture distribution to exceed the uniform-aperture directivity bound $(\pi D/\lambda)^2$ by $+0.14$ to $+0.24$ dB in our 19-feed test case – a real feed array with coupled horns could not do this.
- **Co-polar only.** The radiation integral projects onto the Ludwig-3 co-polar axis only (Eq. (1)); cross-polarization is not computed, so XPD/isolation cannot be assessed with this tool as it stands.
- **One beamforming law, no clustering.** CFM is the only beamforming scheme implemented, and it always runs over the entire feed array – there is no concept of a per-beam feed subset.
- **Idealized feed and single-frequency evaluation.** The $\cos^q \theta'$ taper is an analytic stand-in for a measured feed pattern, and each run is single-frequency: no bandwidth, dispersion, or frequency-scan behavior is modeled.
- **Reflector Geometry.** Only the front-fed, axisymmetric paraboloid is implemented; offset and dual-reflector geometries, and any surface distortion, are out of scope.

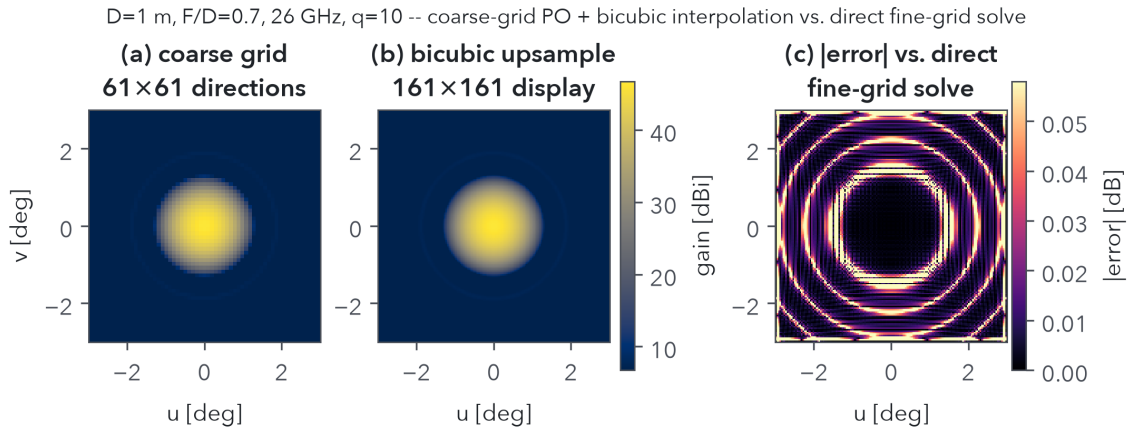


Figure 1: Coarse-grid PO solve (a), bicubic upsample to display resolution (b), and absolute error against a direct fine-grid solve (c), for a single on-axis beam. Colour scale in (a)/(b) spans the top 40 dB of the pattern.

Methodology

Coarse-grid evaluation with complex-field interpolation

A per-beam or composite coverage map requires the field at every pixel of a display grid yet Eq. (1) is bandlimited by the aperture: the far field cannot vary faster than the Nyquist spacing $\delta\theta \approx \lambda/2D$. We therefore solve Eq. (1) directly on a much coarser angular grid, and bicubically (Catmull–Rom) interpolate the *complex* field (not the dB pattern) up to display resolution; dB is sharply nonlinear near the pattern nulls, so interpolating it directly would smear them, while the smooth, bandlimited complex field interpolates almost exactly. Note that inside the PO integral Eq. (1), the term $e^{jk\hat{\mathbf{r}}\cdot\mathbf{r}'}$ oscillates very fast as you sweep across the reflector surface $\mathbf{r}'$. Therefore, it is important to clarify that this is not the axis interpolated. The coarse-to-fine grid is above the observation angle (θ, φ), i.e. it is a function of where you are pointing after the integral has already been evaluated, not a function of the position of the surface.

Fig. 1 demonstrates this on a single on-axis beam (D=1 m, F/D=0.7, 26 GHz, q=10): the field is solved directly on a 61×61 coarse grid (a) and bicubically upsampled to a 161×161 display grid (b); panel (c) is the error against a direct 161×161 PO solve of the same case, evaluated over the region within 40 dB of peak. The 61-point grid gives ~ 8.7 samples across the half-power beamwidth; the resulting error stays below 0.05 dB everywhere within 40 dB of peak (mean 4.8×10^{-3} dB), while solving $61^2/161^2 \approx 7\times$ fewer directions than the direct fine-grid solve. On a full display-resolution coverage map (360×360) we separately measured this saving at $37\times$ wall-clock (Python, 10 cores; see the Results section), consistent with the direction-count ratio. The saving is algorithmic (manifold cost is exactly linear in the number of directions solved) so it stacks with, rather than substitutes for, the language- and threading-level speed-ups reported next.

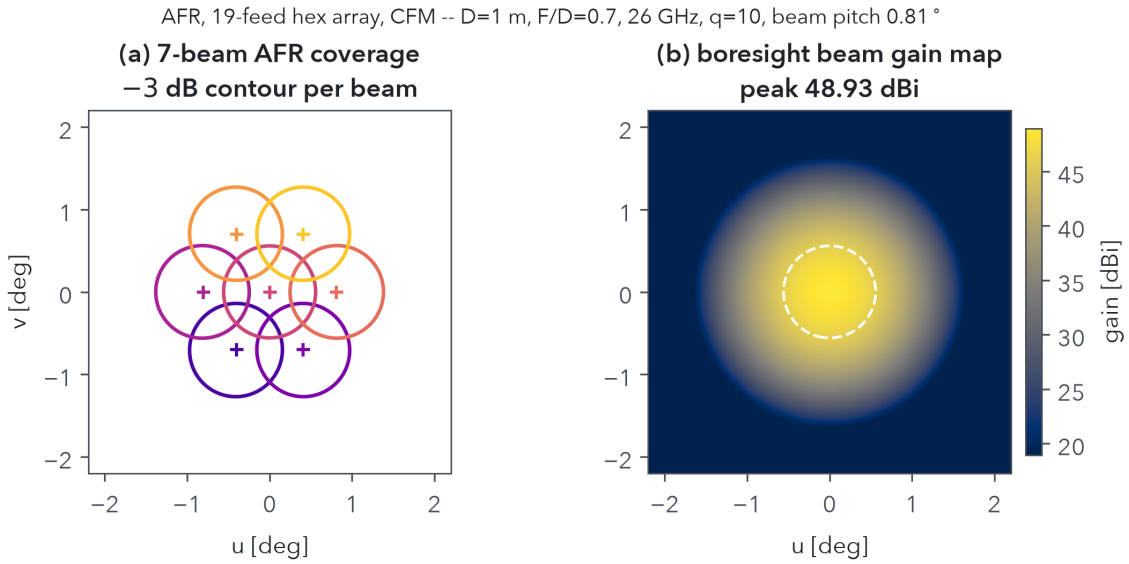


Figure 2: Seven-beam AFR coverage from a 19-feed hexagonal cluster: (a) overlaid -3 dB contour per beam; (b) full colored gain map of the boresight beam, with its own -3 dB contour overlaid (dashed).

Results

Agreement with GRASP

Fig. 2 demonstrates the AFR path on a 19-feed hexagonal cluster (D=1 m, F/D=0.7, 26 GHz, q=10, beam pitch $70\lambda/D \approx 0.81°$, a standard nominal-spacing rule for adjacent spot beams) forming seven simultaneous beams by CFM over the whole array. Panel (a) overlays each beam's own -3 dB contour; panel (b) shows the full colored gain map of the boresight beam (peak 48.93 dBi, in line with the uniform-aperture bound of 48.71 dBi for this D, λ plus the CFM array gain of the 19-feed cluster). All seven beams peak within 0.02 dB of each other, as expected from the hexagonal symmetry of the feed cluster. We emphasize a scope limit here: the GRASP validation below covers the underlying single-feed PO physics only. The CFM combination shown here has been checked for internal consistency between the Rust, Python and JavaScript implementations (agreement $\sim 10^{-13}$ dB, golden cases T4/T5), but AFR beamforming itself has not been independently validated against GRASP.

Fig. 3 compares the model against a GRASP PO run for the on-axis case (D=1 m, F/D=0.7, 26 GHz, q=4.925, edge taper -12 dB) across three cut planes (φ=0°, 45°, 90°, 161 points each). We recomputed this comparison numerically from the underlying GRASP `.cut` data rather than reading values off a plot: peak directivity agrees to **0.031 dB** (47.713 dBi model vs. 47.683 dBi GRASP), and the main beam (within the first null) agrees to within 1.3 dB max / 0.26 dB mean across all three cuts. Point-wise disagreement rises to up to 8 dB in a narrow band immediately around the pattern nulls between the main beam and first sidelobe; there both curves dip at the identical angle sample, so this is a null-*depth* difference (GRASP reaches 3.9 dBi there, the model floors at 7.1 dBi, stable from N=300 to N=600, so it is not a discretization artifact) rather than a location shift, and it is not representative of link-budget-relevant (main-beam) accuracy.

Execution time

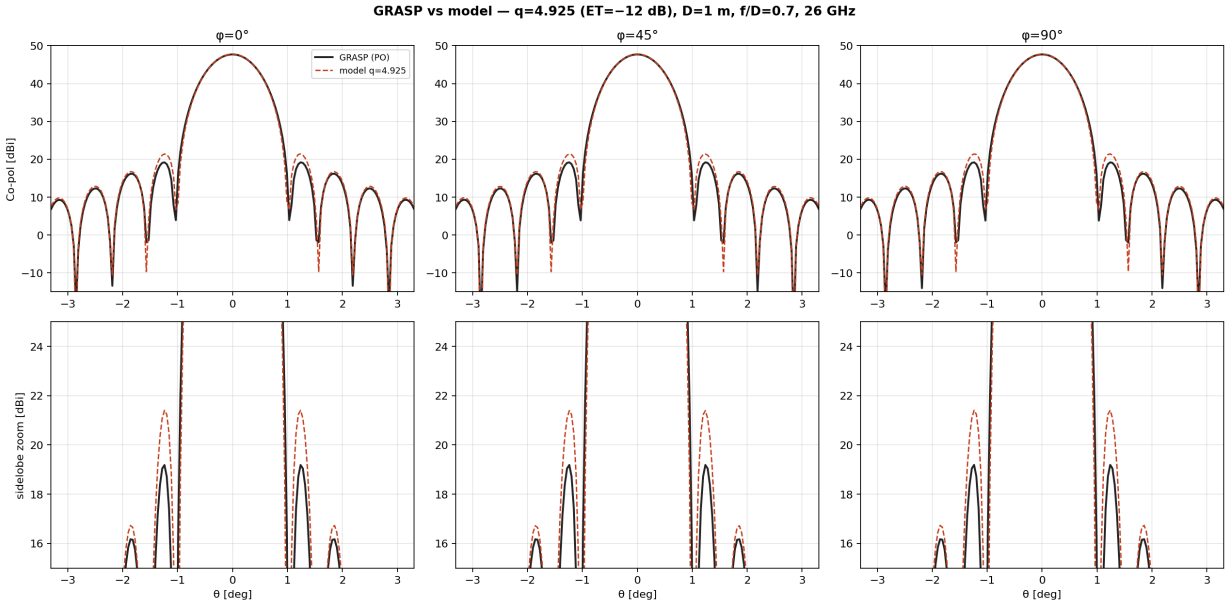


Figure 3: Model (dashed) vs. GRASP PO (solid) co-polar pattern, three cut planes, $D{=}1$ m, $F/D{=}0.7$, 26 GHz, $q{=}4.925$.

Table 1 reports measured wall-clock time for an identical AFR workload (same reflector, direction grid, and feed positions) run independently in Python and Rust, on a 10-core Apple-silicon machine (4 performance + 6 efficiency cores).

Table 1: Wall-clock time, identical AFR workload, M feeds.

M	Python 1 core	Python 10 cores	Rust 1 thread	Rust 10 threads
80	19.75 s	8.67 s	6.32 s	**1.39 s**
200	50.68 s	27.05 s	15.92 s	**3.45 s**

Rust is 3.1–$3.2\times$ faster single-threaded, and 6.2–$7.8\times$ faster using all 10 cores, where Rust's `rayon` scales 4.5–$4.6\times$ against Python `multiprocessing`'s 1.9–$2.3\times$ on the same hardware. Calling the compiled Rust kernel *from* Python through the same C ABI (rather than re-implementing the integral in numpy) recovers pure-Rust performance to within 2–3%: 1.37 s and 3.54 s for $M{=}80, 200$ respectively, all 10 cores parallelized inside the single FFI call. This may be the practical deployment path for teams with an existing Python pipeline (beam layout, caching, configuration, GUI) that need Rust's speed without a rewrite.

The full source, the golden-reference test suite used for every accuracy figure in this paper, and reproduction steps for every timing number in Table 1 are open source, and will be shared shortly at the Laniakea Github repo url.

References

[1] R. J. Acosta, "Active feed array compensation for reflector antenna surface distortions," Tech. Rep. NASA-TM-100826, NASA Lewis Research Center, Cleveland, OH, June 1988.

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BeamCoverage: a tool for GEO coverage and per-beam gain

Laniakea´s **BeamCoverage** allows a system engineer (or similar) to understand **(1)** how many beams you need to cover a target region on ground with a geostationary (GEO) satellite and satisfy system requirements such as total capacity **(2)** visualize the PO-computed per-beam gain on the target region with the performances explained in the results section. The target area can be either specified as a latitude-longitude bounding box (lat_min, lon_min, lat_max, lon_max) or drawn by hand.



Figure 4: European beam coverage with a 4-color scheme

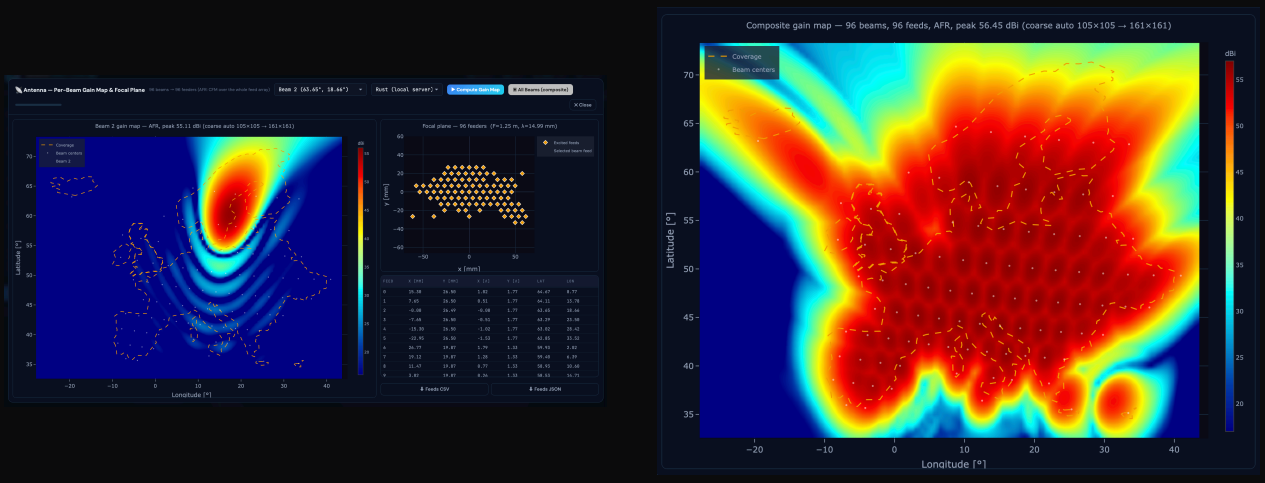


Figure 5: per-beam gain pattern with an AFR antenna using CFM beamforming. Single-beam on the left, full pattern on the right.

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